Laboratory protocol Smallpox Vaccination and CCR5 expression

1) Place falcon-tubes in LAF-bench
2) Pipette 50 µl stain buffer and antibody (see overview on antibodies) to the falcon-tube
3) Pipette 70µl whole blood from the patient to the falcon-tube
4) Vortex 2-3 seconds
5) Incubate for 30 minutes on a tilt-board placed in a dark box.
6) Add 1.25ml BD Lysing Solution (Must be room temperature)
7) Vortex 2-3 seconds
8) Incubate for 20 minutes on a tilt-board placed in a dark box.
9) Centrifuge falcon-tube med blood and BD Lysing Solution at 300G in 5 minutes.
10) Extract the liquid from the Falcon-tube
11) Add 3ml PBS and resuspend.
12) Centrifuge at 300G in 5 minutes.
13) Extract liquid from the Falcon-Tube and leave 200 µl in the tube
14) Analyze on Flow Cytometer. 400 µl is read by flow cytometer.

Dilution factor must be considered when cell counting. According the manufacture of Transfix, the absolute number must be multiplied by 1.2.

Transfix-tubes must be stored in temperature between 2 and 8 degrees celcius.
Before analyzing of the results an antibody compensation must be performed.

BD Lysing Solution

- use 1ml BD Lysing Solution per 50 µl whole blood.
- Is mixed in ratio 1:10
  - 1 ml BD Lysing Solution + 9ml sterile water
- Must be mixed in chemical fume hood.
- When adding BD Lysing Solution to whole blood the mixture must incubate for 20 minutes.

Antibodies

Antibody	FITC	PE	PECF594	PerCp cy5.5	APC-H7	BV421	BV605	BV786
Chemokine	CD3	CCR7	CD28	CD8	CD4	CCR5	CD27	CD45RA